

Formal (Lean kernel)

AQFT core

Breuer corner

four-frame

solder field

Einstein emerge

transport witness

Physics (certificates)

spin-2 massless

2 helicities

ghost-free

conservation

no anomalies

Experiment (V11 rite)

unblinded

mocks

systematics

powered verdict

TGL_QG_MODEL_FORMALLY_CLOSED__NATURE_TEST_COMPLETED_WITHIN_LOCAL_BULK_AT_AVAILABLE_SENSITIVITY__MORE_SENSITIVE_DATA_COULD_REVERSE

fail-closed gate: each box flips ONLY by construction (Lean term + clean axioms) or by a pre-registered powered verdict; NOT_FALSIFIED $\neq$ CONFIRMED